

Point cloud-based morphological analysis and surface reconstruction of extraterrestrial lava tubes

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Abstract

Lava tubes provide vast and structurally stable underground spaces with minimal temperature fluctuations and effective shielding from cosmic radiation, making them promising candidates for extraterrestrial base construction. Assessing their feasibility requires detailed morphological analysis and high-precision three-dimensional surface reconstruction. This paper proposes a point cloud-based surface reconstruction method. Principal Component Analysis (PCA) is used for adaptive normal estimation and normalization, while an anisotropic weighting function combined with the Moving Least Squares (MLS) projection is used to resample curvature-sensitive regions. An implicit surface is then reconstructed using the Poisson algorithm to generate a high-quality triangular mesh. The proposed method is validated using an Indian lava tube dataset. Twenty-seven typical cross-sections are uniformly extracted along the central axis of the lava tube constructed by the uniform drift algorithm, with their morphological characteristics analyzed. After that, ten point cloud regions with typical features of the lava tube were extracted for surface reconstruction and error analysis. The effect of the initial neighborhood radius (30–80 mm) on normal vector estimation was investigated. The results indicate that the geometric similarity error of point cloud resampling can be minimized to 2 mm, which indicating that our method can reconstruct a surface with high accuracy and preserve local structural details. Furthermore, it provides a theoretical basis and methodological support for the site selection of extraterrestrial bases.

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1. Introduction

As one of the research focuses in extraterrestrial exploration, lava tubes can provide vast and structurally stable

underground space with low temperature difference and cosmic radiation environment (Léveillé and Datta, 2010), which makes the best choice to build up underground extraterrestrial habitats in the future (Mari et al., 2021). Lava tubes are widely distributed on Earth, and lava-tube-like voids have also been identified on the Moon and Mars. Their morphological characteristics, spatial distribution, and material composition not only have significant scientific value for understanding the internal evolutionary history of these planets but also play an

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important role in supporting future extraterrestrial human activities, including transportation, infrastructure development, and habitat construction (Titus et al., 2021).

In the study of extraterrestrial lava tubes, geometric parameters such as height, width, and cross-sectional morphology are of critical importance for understanding their formation processes and evaluating their potential for underground habitats (Theinat et al., 2020; Theinat et al., 2018). Recent studies on lunar lava tubes have shown that irregular morphological characteristics are key factors affecting their stability (Chwała et al., 2024). Therefore, the cross-sectional geometry must be considered to accurately assess the stability of the roof. These parameters are not only the basis for three-dimensional modeling but also important references for path planning, environmental simulation, and remote operations (Patel et al., 2023; She et al., 2025).

Current approaches for detecting extraterrestrial lava tubes include gravity measurements, optical imaging, and radar detection. Gravity satellites can determine the distribution of lava tubes using anomalous signals; optical images can identify surface collapse openings using two-dimensional images (Haruyama et al., 2009); and ground-penetrating radar actively obtains echo signals and determines the upper and lower interfaces of lava tubes by recognizing image textures. However, these methods are mainly indirect inferences and two-dimensional approximations, with low measurement accuracy, making it difficult to gather structural information directly from lava tubes. To overcome the limitations of long-distance, non-contact remote sensing approaches, using LiDAR to enter lava tubes and collect three-dimensional point cloud data is critical for future extraterrestrial exploration. LiDAR placed on a robotic platforms may actively scan inside tunnels, capturing high-density, high-precision three-dimensional point clouds of interior surfaces. The point cloud data accurately depicts the geometry of the lava tubes in three dimensions, displaying cross-sectional contours and top-wall undulations and giving credible data for structural stability assessment and surface reconstruction.

However, during LiDAR acquisition of point clouds, the data are often affected by multipath interference, high-reflectivity surfaces, sensor noise, and hardware limitations, resulting in noisy points, outliers, and artifacts (Berger et al., 2017). These disturbances can significantly degrade the quality of subsequent surface reconstruction, manifesting as voids, geometric distortions, or surface irregularities. To obtain smooth, continuous, and geometrically faithful mesh models, it is essential to perform point cloud resampling to mitigate the impact of data deviations on surface reconstruction. Various methods have been proposed for resampling 3D point clouds (Alexa et al., 2003; Ding et al., 2023). The Local Optimal Projection algorithm enables iterative projection of point cloud subsets without requiring parameter tuning (Lipman et al., 2007). With the development of deep learning, a neural implicit field-based point cloud upsampling approach has been proposed

that transforms sparse point clouds into dense and uniformly distributed representations, thereby enhancing structural completeness (Li et al., 2023). Additionally, an entropy-maximizing gradient flow-based downsampling method has been introduced that can preserve geometric information while downsampling (Petracek et al., 2024).

To preserve fine structures in high-curvature areas and achieve smooth transitions in low-curvature regions, thereby enhancing reconstruction accuracy and stability in complex terrain conditions, this study builds upon Optimized Surface Reconstruction (OSR) method (Yao et al., 2025) by introducing an adaptive neighborhood normal vector estimation approach. Because there is currently a paucity of true high-resolution point cloud data for extraterrestrial exploration, this study demonstrates the applicability of the method with a 3D point cloud dataset of an Indian lava tube on Earth, serving as a methodological reference for future extraterrestrial exploration.

This paper proposes a geometric morphology analysis framework and a point cloud-based Adaptive Surface Reconstruction (ASR) method for extraterrestrial lava tubes, aiming to provide technical support for the remote sensing identification and geometric modeling of extraterrestrial lava tubes. The structure of the paper is organized as follows: Section 2 describes the proposed methodology, including centerline extraction of lava tubes, point cloud resampling strategies and a curvature-optimized 3D surface reconstruction algorithm. Section 3 defines the evaluation metrics for resampling and surface reconstruction. Section 4 presents experimental results with public datasets, accompanied by detailed analysis and discussion. Section 5 concludes the study and outlines potential future research directions.

2. Proposed method

2.1. Lava tube morphological analysis

The morphological analysis of the irregularly shaped lava tube proceeds as follows: First, the original point cloud is voxel-downsampled and segmented along the initial axis of the lava tube to obtain the initial seed points for Mean Shift (MS) processing. Then, the MS algorithm is applied to extract an initial set of centerline points. A cubic spline interpolation (McKinley and Levine, 1998) is performed to fit a continuous principal axis curve, along which uniformly spaced samples are taken and locally refined to improve the accuracy of the estimated centerline. Finally, cross-sections are generated at equal intervals along the centerline. For each section, the corresponding point cloud is projected onto a local 2D plane, and the convex hull algorithm (Barber et al., 1996) is used to extract the cross-sectional contour and compute geometric descriptors.

The MS algorithm performs an iterative search over sample points in Euclidean space, guiding each search point in the direction of increasing density until it

converges to a local maximum of the underlying density function (Carreira-Perpinán, 2015). Let the initial point cloud obtained after downsampling and segmentation be denoted as $X = \{x_1, x_2, \dots, x_n\}$. The Gaussian kernel function is defined as:

$$K(u) = \exp\left(-\frac{1}{2}\|u\|^2\right) \quad (1)$$

For any point $x \in X$, the kernel density estimation is defined as:

$$\hat{f}(x) = \frac{1}{nh^d} \sum_{i=1}^n K\left(\frac{x - x_i}{h}\right) \quad (2)$$

Here, $d = 3$ indicates that the initial point cloud resides in a three-dimensional Euclidean space. The bandwidth h serves as the search radius, which determines the neighborhood range involved in the weighted calculation for each iteration.

To simplify the expression of the gradient, the Gaussian kernel is written in the form $K(u) = k(\|u\|^2)$, and define $g(s) = -k'(s)$. By differentiating the kernel density estimation, the gradient of the estimated density is obtained as:

$$\nabla \hat{f}(x) = -\frac{1}{nh^d} \sum_{i=1}^n (x - x_i) \cdot g\left(\left\|\frac{x - x_i}{h}\right\|^2\right) \quad (3)$$

Eq. 3 indicates that the gradient direction of the kernel density is determined by the relative positions of neighbor points with respect to the initial search point, while the magnitude is governed by the derivative of the kernel function. Therefore, $g(\cdot)$ is treated as a weight function to perform a weighted average over the sample points within the search radius of the initial point x . The difference between this weighted average and x yields a displacement vector that points toward the direction of increasing local density, known as the MS vector, which is defined as:

$$m(x) = \frac{\sum_{i=1}^n x_i \cdot g\left(\left\|\frac{x - x_i}{h}\right\|^2\right)}{\sum_{i=1}^n g\left(\left\|\frac{x - x_i}{h}\right\|^2\right)} - x \quad (4)$$

The formula for computing the next search point in the iterative process of MS is defined as:

$$x^{(t+1)} = x^{(t)} + m(x^{(t)}) = \frac{\sum_{i=1}^n x_i \cdot g\left(\left\|\frac{x^{(t)} - x_i}{h}\right\|^2\right)}{\sum_{i=1}^n g\left(\left\|\frac{x^{(t)} - x_i}{h}\right\|^2\right)} \quad (5)$$

The iteration continues until the constraint condition $|x^{(t+1)} - x^{(t)}| < \varepsilon$ is satisfied, where ε denotes the convergence threshold.

Fig. 1 illustrates the three-step iterative process of the MS algorithm. S_1 and S_2 represent kernel windows constructed with a search radius h in continuous iterations, which define the local sample regions involved in the

current mean estimation. The point x_1 denotes the initial search location. By computing the weighted average of the neighbor samples within S_1 , the search point moves along the gradient direction of the local density to a new position x_2 . Further iteration yields x_3 , with which the search point gradually converging toward the region of highest local density. Since the lava tube segments exhibit a hollow structure with annular point cloud distributions, the MS algorithm progressively shifts the search point toward the geometric center with the highest kernel density, thereby enabling the construction of the central axis.

2.2. Point cloud resampling

2.2.1. Normal vector estimation

In point cloud processing, point normal estimation is crucial for feature extraction, surface reconstruction, and 3D reconstruction. To improve the accuracy of normal estimation under different curvature conditions, this paper proposes an adaptive normal vector estimation method based on Principal Component Analysis (PCA). This method dynamically adjusts the neighborhood scale according to local geometric features, thereby maintaining smoothness in flat areas while preserving detail accuracy in high-curvature regions. By fitting a local plane within the adaptive neighborhood and using its normal vector as the estimated normal vector for the sampled points, the method can effectively enhance the robustness and consistency of overall point cloud normal estimation.

Let the point cloud be denoted as $P = \{p_1, p_2, \dots, p_n\}$. For any point p_i , an initial neighborhood radius r_0 is used to select its neighboring points, forming the neighbor set $Q_i = \{q_1, q_2, \dots, q_k\}$. To remove the influence of the global coordinate system and mitigate systematic bias from curvature, each neighboring point $q_j \in Q_i$ is transformed into a local coordinate system centered at the neighborhood centroid μ_i :

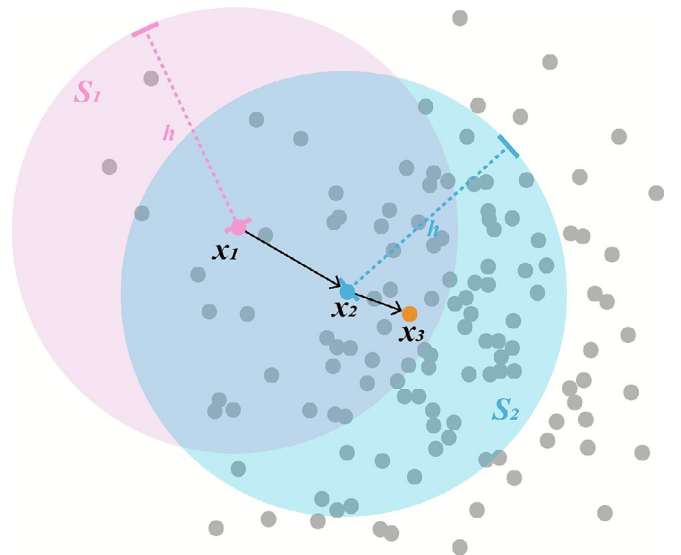


Fig. 1. MS algorithm iteration principle diagram.

$$q'_j = q_j - \mu_i \quad (6)$$

The covariance matrix of the neighborhood is then computed as

$$C_i = \frac{1}{k-1} \sum_{j=1}^k q'_j (q'_j)^T \quad (7)$$

The covariance matrix C_i is a symmetric 3×3 matrix, and its eigenvalues can be computed by

$$C_i v_{i,m} = \lambda_{i,m} v_{i,m}, \quad m = 1, 2, 3 \quad (8)$$

Here, $\lambda_{i,m}$ and $v_{i,m}$ denote the m -th eigenvalue and the corresponding eigenvector, respectively. The eigenvalue $\lambda_{i,m}$ represents the dispersion of the local point distribution along the direction $v_{i,m}$, with $\lambda_{i,1} \leq \lambda_{i,2} \leq \lambda_{i,3}$. The eigenvector corresponding to the smallest eigenvalue, $v_{i,1}$, represents the direction of least variation within the neighborhood of p_i , and $n_i^{(0)} = v_{i,1}$ is taken as the initial normal vector of point.

To make the normal vector sensitive to curvature and adaptive to different geometric regions, a curvature-sensitive metric based on the PCA eigenvalues, denoted as s_i :

$$s_i = \frac{\lambda_{i,1}}{\lambda_{i,1} + \lambda_{i,2} + \lambda_{i,3}} \quad (9)$$

A smaller value of s_i indicates that the point and its neighborhood are approximately planar, whereas a larger s_i implies that the neighborhood exhibits higher curvature or boundary features. According to the magnitude of s_i , the algorithm adjusts the neighborhood radius adaptively based on a Sigmoid curvature response function, which is defined as:

$$\text{Sigmoid}(s_i) = \frac{1}{1 + e^{-(s_i - s_c)}} \quad (10)$$

where s_c is the center point of the Sigmoid function. Based on the Sigmoid response function, the adaptive neighborhood radius r_i is calculated as:

$$r_i = r_0 \cdot [(1 - \eta) + (\gamma + \eta) \cdot (1 - \text{Sigmoid}(s_i))] \quad (11)$$

Here, γ and η denote the neighborhood expansion and contraction coefficients (typical values: $\gamma = 0.2, \eta = 0.3$), and r_0 is the initial neighborhood radius. When s_i is small, the algorithm increases the neighborhood radius appropriately to enhance the stability and noise robustness of the normal vector estimation. When s_i is large, the neighborhood radius is reduced to prevent excessive averaging across different surfaces. In the transition region, the neighborhood radius varies continuously according to the curvature response function, allowing the normal vector estimation to change smoothly between different geometric regions, thereby preserving structural details while improving the overall continuity and numerical stability of the computation.

Subsequently, the point p_i reselects its neighborhood set Q'_i based on the adjusted radius r_i and the normal vector n_i of the point is computed accordingly.

2.2.2. Calculation of anisotropic weights

$n_1 = (x_1, y_1, z_1)$ and $n_2 = (x_2, y_2, z_2)$ are the unit normal vectors at p_1 and p_2 , respectively. α is angle of between n_1 and n_2 , denoted by follow Equation.

$$\alpha = 2 \arcsin \left(\frac{\|n_1 - n_2\|}{2} \right) \quad (12)$$

The normal vector distance N_d between them as follows:

$$\begin{aligned} N_d &= \|n_1 - n_2\| \\ &= \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2} \\ &= 2 \sin \left(\frac{\alpha}{2} \right) \end{aligned} \quad (13)$$

The traditional MLS method applies isotropic weighting function based solely on the Euclidean distance between points but neglecting the important geometric feature of point normals. This often leads to excessive smoothing in regions with sharp features. To solve this issue, the proposed method introduces the normal vector distance as a metric for geometric difference and propose a normal-difference-based weighting strategy. Specifically, for any point p , if the normal distance of p and its neighbor point does not exceed a predefined threshold, it will be incorporated into the fitting process with a weight determined by a Gaussian function. Otherwise, it is excluded. The weighting function with normal distance constraint is defined as:

$$\theta(d) = \begin{cases} e^{-d^2/r^2}, & \text{if } N_d \leq \xi \\ 0, & \text{if } N_d > \xi \end{cases} \quad (14)$$

Where d is the Euclidean distance between point p and its neighbor point. ξ is threshold of normal vector distance. r is the neighborhood radius. θ is weight function. θ is used to optimize the contribution of neighbor points in fitting process. When the normal vector distance N_d is below the ξ , the weight of the neighbor points decreases with the increasing distance.

2.2.3. Point cloud reprojection

This study adopts a point resampling method based on projection algorithm to optimize the point cloud distribution by performing local fitting and adjustment for each point.

The procedure is as follows: Let the original point cloud be denoted as $P = \{p_1, p_2, \dots, p_n\}$ and construct a KD-Tree on P to accelerate neighborhood queries. For each point $p \in P$, a point set $T = \{p, q_1, q_2, \dots, q_i, \dots, q_m\}$ is established. q_i are the neighbor points within the radius of the neighborhood of the p .

A plane fitting algorithm is applied in estimating the supporting plane H with point set T of each point p . The equation of H is give as follows:

$$Ax + By + Cz + D = 0 \quad (15)$$

The parameters A, B, C, D of the supporting plane H can be determined by minimizing the sum of the squares of the weighted projection distances from the neighbor points $q_i \in T$ to the plane, as follows:

$$\min \sum_{i=1}^m (\langle n, q_i \rangle + D)^2 \theta(\|q_i - p\|) \quad (16)$$

$n = (A, B, C)$ is the normal vector of the supporting plane H and $\langle n, q_i \rangle + D$ represents the signed distance from q_i to H . $\theta(\|q_i - p\|)$ is a weighting function introduced in Section 2.2.2.

The point cloud often exhibits nonlinear distributions in practical applications, but the supporting plane H can capture the linear features in the neighborhood of point p . As a first-order approximation model, the plane H cannot accurately represent the true geometric shape. To capture more nonlinear characteristics of the point cloud, the fitted surface g is obtained from the point set T using the least squares method, as follows:

$$g(X) = C \varphi^T(X) = \sum_{j=1}^{10} c_j \varphi_j \quad (17)$$

Here, $X(x, y, z)$ represents a point in 3D space. $\varphi(X)$ is the vector of basis functions used for polynomial surface fitting, defined as follows:

$$\varphi(X) = (1, x, y, z, xy, xz, yz, x^2, y^2, z^2) \quad (18)$$

The corresponding associated coefficient vector is C :

$$C = (c_1, c_2, c_3, c_4, c_5, c_6, c_7, c_8, c_9, c_{10}) \quad (19)$$

Establish a local Cartesian coordinate system with point p as the origin and the normal vector n of H as the z -axis. To determine the coefficient vector C that allows the fitted surface g to best approximate the point set T , the weighted sum of squared residuals between predicted results and ground truth is minimized. The objective function as follows:

$$J = \sum_{i=1}^m \theta(\|q_i - p\|) \cdot [g(q_i) - n \cdot (q_i - p)]^2 \quad (20)$$

Here, $g(q_i)$ denotes the predicted value of the fitted surface at q_i . The term $n \cdot (q_i - p)$ represents the signed distance from q_i to point p along the normal direction n , which serves as the observed value. $\theta(\|q_i - p\|)$ is a weighting function.

To minimize the objective function J , take the partial derivative of Eq. 20 with respect to each coefficient c_j and set it to zero, as follows:

$$\frac{\partial J}{\partial c_j} = 2 \left\{ \sum_{i=1}^m \theta(\|q_i - p\|) \left[\sum_{k=1}^{10} c_k \varphi_k(q_i) - n \cdot (q_i - p) \right] \varphi_j(q_i) \right\} = 0 \quad (21)$$

Eq. 20 presents a set of equations with ten unknowns, that is the coefficient c_j . The solution to this set corresponds to

the optimal coefficient vector C that best fits the surface. To further simplify the expression and facilitate numerical computation, the system can be expressed in matrix form as follows:

$$\frac{\partial J}{\partial C} = 2A^T(AC - b) = 0 \Rightarrow AC = b \quad (22)$$

The matrix A defined in the Eq. 23 is formed by the weight function and the basis function of the points in set T .

$$A = \begin{bmatrix} \sqrt{\theta(\|q_1 - p\|)} \varphi_1(q_1) & \cdots & \sqrt{\theta(\|q_1 - p\|)} \varphi_{10}(q_1) \\ \sqrt{\theta(\|q_2 - p\|)} \varphi_1(q_2) & \cdots & \sqrt{\theta(\|q_2 - p\|)} \varphi_{10}(q_2) \\ \vdots & & \vdots \\ \sqrt{\theta(\|q_m - p\|)} \varphi_1(q_m) & \cdots & \sqrt{\theta(\|q_m - p\|)} \varphi_{10}(q_m) \end{bmatrix} \in \mathbb{R}^{m \times 10} \quad (23)$$

The Eq. 24 defines the weighted observation vector b .

$$b = \begin{bmatrix} \sqrt{\theta(\|q_1 - p\|)} n \cdot (q_1 - p) \\ \sqrt{\theta(\|q_2 - p\|)} n \cdot (q_2 - p) \\ \vdots \\ \sqrt{\theta(\|q_m - p\|)} n \cdot (q_m - p) \end{bmatrix} \in \mathbb{R}^m \quad (24)$$

Then the coefficient vector C can be represented by matrix S and the vector b .

$$C = (A^T A)^{-1} A^T b \quad (25)$$

The point projection process is illustrated in Fig. 2. After fitting the supporting plane and the surface, for each point in the point set P , find its intersection of the normal vector n and the fitting surface g . These so-called projection points or the resampling points form the projection point set $R = \{r_1, r_2, \dots, r_n\}$. The size of the point cloud remains unchanged after resampling, but the positions of the points are altered.

2.2.4. Principle of Poisson reconstruction

Poisson reconstruction (Kazhdan et al., 2006) is an implicit surface reconstruction method for point clouds. It involves obtaining the gradient of the indicator function from an oriented point set, solving for the indicator function, and then extracting the isosurface to obtain the surface model. The indicator function $\tilde{\chi}$ is derived by solving the Poisson equation.

In the three-dimensional Cartesian coordinate system, the Poisson equation is expressed as follows, where φ and f represent the real-valued and complex-valued functions on the manifold, respectively.

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) \varphi(x, y, z) = f(x, y, z) \quad (26)$$

Given a set of point cloud samples, each sample consists of a point and its normal vector. To achieve watertight Poisson surface reconstruction, the key issue is to compute the indicator function from the input point cloud samples. Starting from the relationship between the gradient of the

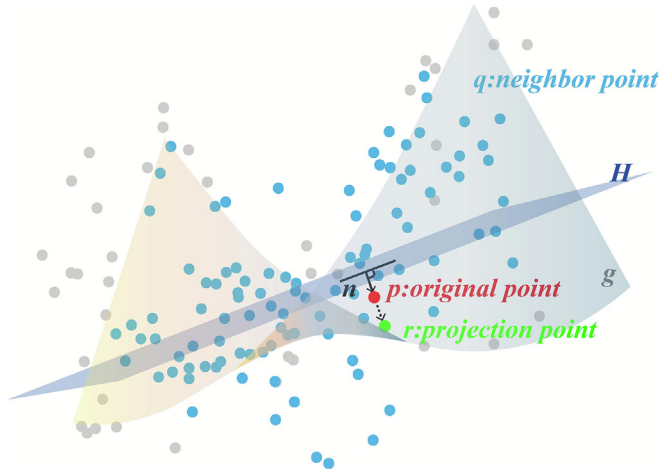


Fig. 2. Projection of a point along the support plane normal onto the fitted surface.

indicator function and the surface normal field, the gradient field is derived from point cloud samples with normals, and the indicator function is obtained from this gradient field.

First, the gradient field is defined. The indicator function was a piecewise constant function. An accurate calculation of the gradient field results in infinite vector field values at the surface boundaries. To prevent this uncontrollable situation, a smoothing filter is used to convolve the indicator function, and the gradient field of the smoothing function is then used. The smoothed indicator function gradient field equals the vector field.

After obtaining the vector field $\vec{V}$, the next step is to solve for function $\tilde{\chi}$. A Poisson equation was constructed to find the optimal least-squares approximate solution.

$$\Delta \tilde{\chi} = \nabla \cdot \vec{V} \quad (27)$$

Here, Δ is the Laplacian operator, and ∇ is the divergence operator. The gradient of the indicator function is essentially a vector field, whose value is zero on the surface of the model and non-zero at other locations.

The surface generated by Poisson reconstruction has the characteristic of being watertight, with a smooth surface and strong robustness.

3. Evaluation metrics

3.1. Point cloud resampling

3.1.1. Geometric similarity

To evaluate the geometric consistency between the resampled point cloud and the original point cloud, we use a point cloud similarity metric as the quality assessment indicator for resampling. Let the original point cloud set be $P = \{p_1, p_2, \dots, p_m\}$ and the resampled point cloud set be $R = \{r_1, r_2, \dots, r_n\}$, where p_j and r_i denote the 3D coordinates of points in the point clouds. The basic idea of the

point cloud similarity metric is to calculate the spatial geometric distance between R and P . A smaller distance indicates a higher similarity in shape and distribution between the two point cloud sets. The Root Mean Square error (RMSE) between the resampled and original point clouds is defined as follows:

$$\text{RMSE}(R, P) = \sqrt{\frac{1}{n} \sum_{i=1}^n \min_{p_j \in P} \|r_i - p_j\|^2} \quad (28)$$

where $\|r_i - p_j\|$ denotes the Euclidean distance between the resampled point r_i and the original point p_j . A smaller RMSE indicates that the resampled point cloud is closer to the original point cloud in terms of spatial distribution and geometric shape, implying higher resampling quality.

3.1.2. Curvature similarity

To evaluate the preservation of geometric features, the differences in local curvature distributions between the original and resampled point clouds are compared. First, the Mahalanobis distance-based curvature κ is computed for each point to represent the local surface variation. Then, a KDTree is employed to establish point-to-point correspondences between the resampled and original point clouds. Finally, the mean absolute curvature difference $|\Delta\kappa|$ of the matched pairs is calculated. This metric effectively reflects the fidelity of local surface details during the resampling process, where smaller values of $|\Delta\kappa|$ indicate better preservation of curvature characteristics.

3.2. Surface reconstruction

The Hausdorff distance is a commonly used metric for quantifying the dissimilarity between two non-empty subsets in a metric space (Huttenlocher and Kedem, 1990). It is defined as the maximum deviation between each point in one set to the nearest point in the other set. For a metric space (M, d) , and two non-empty subsets $X, Y \subset M$, $d_H(X, Y)$ is introduced to represent the Hausdorff distance between two sets and its definition is given by the following formula:

$$d_H(X, Y) = \max \left\{ \sup_{x \in X} d(x, Y), \sup_{y \in Y} d(y, X) \right\} \quad (29)$$

where $d_H(x, Y)$ and $d_H(X, y)$ both represent the point-to-set distance. Furthermore, the point-to-point distance can be defined by the distance of two points. Eq. 30 illustrates the relationship between them, while the $d(a, b)$ denotes the distance between two points.

$$d(a, B) := \inf_{b \in B} d(a, b) \quad (30)$$

In this study, the Hausdorff distance is used to evaluate the geometric accuracy of reconstructed surfaces by measuring the deviation between the point cloud and the generated mesh. A smaller Hausdorff distance indicates a higher fidelity of the reconstructed surface in approximating the

original geometry. Here, two point sets are the original point cloud $P = \{p_1, p_2, \dots, p_i, \dots, p_n\}$ and the set $S = \{s_1, s_2, \dots, s_j, \dots, s_m\}$ formed by the triangular mesh. Each mesh is defined by its three vertices as $s_j = (q_0, q_1, q_2)$. The Hausdorff distance list $d(P, S)$ between the two sets represents the error of the reconstruction result. The process of calculating the reconstruction error is as follows:

$$n_j = \frac{(q_1 - q_0) \times (q_2 - q_0)}{\|(q_1 - q_0) \times (q_2 - q_0)\|} \quad (31)$$

Compute the distance $d(p_i, s_j)$ from p_i to the triangular face s_j as follows:

$$d(p_i, s_j) = \begin{cases} \|p_i - p_{\text{proj}}\|, & \text{if } p_{\text{proj}} \in s_j \\ \min\{d(p_i, e_j)\}, & \text{otherwise} \end{cases} \quad (32)$$

Here, p_{proj} denotes the orthogonal projection of p_i onto the plane of triangle s_j along its normal vector n_j .

$$p_{\text{proj}} = p_i - [(p_i - q_0) \cdot n_j] n_j \quad (33)$$

e_j represents each point along the edge of the triangle face s_j . Then the distance $d(p_i, S)$ from point p_i to the mesh S can be computed with Eq. 30.

After that, the mean error (ME) and root mean square error (RMSE) of the reconstructed surface are introduced in this paper as evaluation metrics for surface reconstruction. The ME of the reconstructed surface is given by:

$$ME = \frac{1}{n} \sum_{i=1}^n d(p_i, S) \quad (34)$$

The RMSE is given by:

$$RMSE = \frac{1}{n} \sum_{i=1}^n (d(p_i, S) - ME)^2 \quad (35)$$

Here, n denotes the number of points in the point cloud. The ME reflects the overall average deviation, while the RMSE is more sensitive to local errors and can capture the surface smoothness and uniformity of error distribution. Smaller values of both indicate that the reconstructed surface better fits the original point cloud, implying higher geometric accuracy.

4. Data analysis and test results

4.1. Dataset

To validate the proposed morphological analysis and surface reconstruction methods for Martian lava tubes (Quan et al., 2016), we employed the planetary pit and cave simulation dataset provided by the National Aeronautics and Space Administration (NASA). This dataset contains dense 3D point clouds from four real-world environments. Among them, the "Indian Tunnel" is acquired with high-resolution scan of a lava tube interior, consisting of approximately 110 million points and exhibiting some noise. Fig. 3 presents the point cloud in four forms: raw,

RGB-colored, height-based and density-based representation.

Table 1 presents ten selected feature regions from the point cloud dataset, including their point counts and visualizations from different perspectives (front, top, and right views). Seg0 to Seg9 correspond to the wall-ground junction area, left sidewall, right sidewall, wall protrusion-transition area, top protrusion area, top depression area, top flat area, flat ground area, gravel-covered ground area and fallen-rock ground area, respectively.

4.2. Lava tube morphological analysis

The total length of the Indian lava tube is approximately 250 meters. To intuitively characterize the geometric variations along its axial direction, a centerline extraction algorithm was first applied in the 3D point cloud data to obtain the central axis curve representing the primary orientation of the lava tube, as shown in Fig. 4. (see Fig. 5).

Along the extracted centerline, the sampling interval was dynamically adjusted based on local curvature. At each sampling position, a normal plane was constructed to intersect with the point cloud, resulting in 27 representative cross-sectional point sets, as shown in Fig. 4. cross-sections along with their corresponding measurement parameters.

To eliminate the influence of noise points on geometric measurements, a statistical outlier removal method with a standard deviation parameter of 1.2 was first applied to filter out abnormal values. Subsequently, PCA was used to construct a local coordinate system for each cross-section. The point cloud was then projected onto a two-dimensional plane defined by the tunnel's longitudinal (ground) and vertical directions. Using a 2D convex hull algorithm, the boundary of each cross-section was extracted to achieve a closed contour. It should be noted that the convex hull algorithm simplifies the complex cross-sectional contours of the lava tubes into minimal convex polygons. Consequently, it cannot capture internal concave structures, systematically overestimating the cross-sectional area. The extent of this overestimation varies with the geometry of each section. Despite this limitation, the convex hull algorithm offers advantages in maintaining contour continuity and computational robustness, providing a consistent reference for the morphological analysis in this study. Therefore, the cross-sectional area values reported here represent an upper-bound estimate of the practically usable area.

Based on the above processing, four key geometric parameters of 27 cross-sections including width (along the ground direction), height (vertical direction), area and aspect ratio can be obtained. The Fig. 6 shows the boundary fitting results of several representative cross-sections along with their corresponding measurement parameters. The overall shape appears as an irregular arch.

The morphological parameters of different cross-sections are compared in the following analysis and the

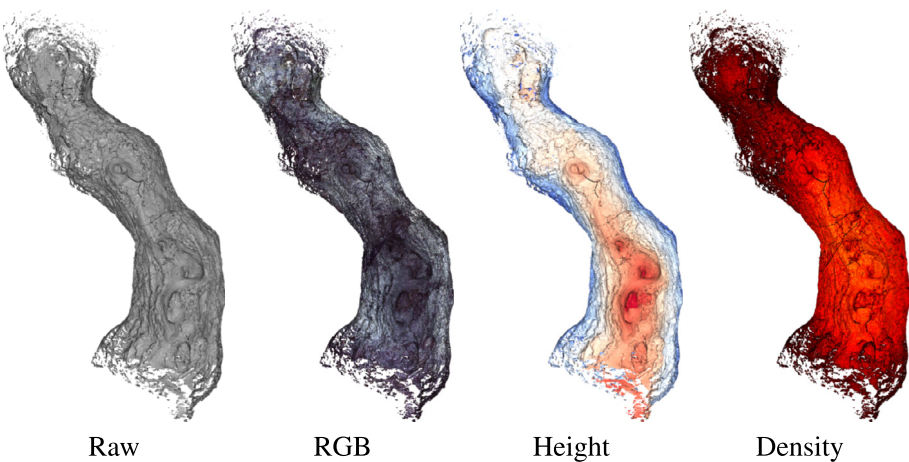


Fig. 3. Indian Tunnel point cloud shown in raw, RGB, height, and density-based representation.

Table 1
Multi-view visualizations of Seg0 to Seg9 feature regions.

Feature	Multi-view			Feature	Multi-view		
Seg0 (40511)				Seg5 (56544)			
Seg1 (70728)				Seg6 (42377)			
Seg2 (62056)				Seg7 (61077)			
Seg3 (47615)				Seg8 (75938)			
Seg4 (84385)				Seg9 (86957)			

results are shown in the Fig. 7. The height of the lava tube ranges from 6.39 to 12.83 m, showing an overall trend of increasing followed by decreasing. The width ranges from 12.01 to 24.55 m with significant fluctuations, generally exhibiting a pattern of being wide at both ends and narrow in the middle. The aspect ratio varies between 1.25 and 3.7, indicating that some cross-sections are relatively elongated. The cross-sectional area between 63 and 169 m² is influ-

enced by both width and height. Morphological analysis reveals that geometric parameters vary considerably along the lava tube. However, within the cross-sections between 4 and 7, the variations in all four key parameters are relatively small, indicating a more regular and stable structure in this region.

The regular geometric morphology is only one of the key factors in evaluating lava tubes as potential extraterres-



Fig. 4. Extracted centerline of the Indian Tunnel lava tube.



Fig. 5. Representative cross-sections sampled along the extracted centerline.

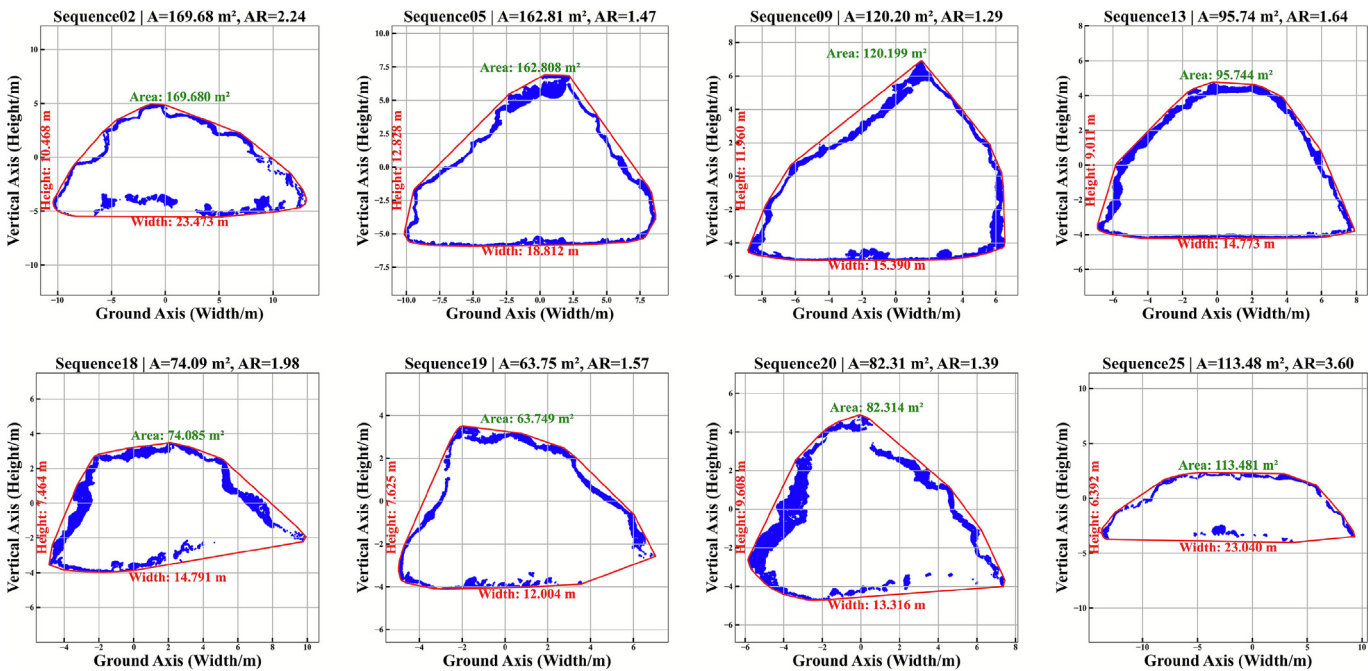


Fig. 6. Cross-sectional profiles of lava tube, the red curve represents the fitted contour line.

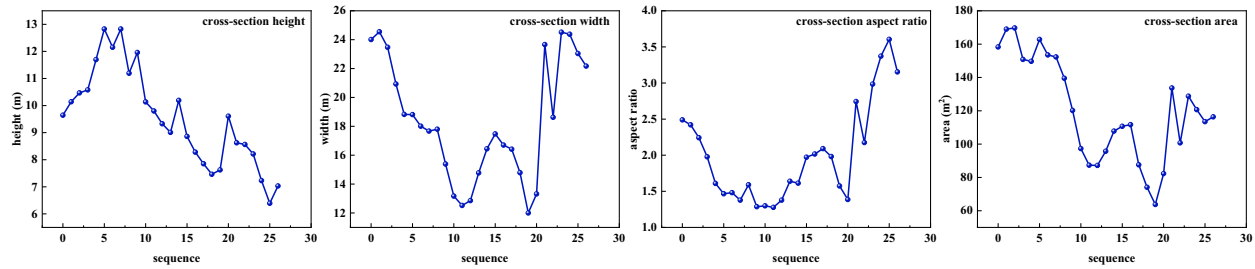


Fig. 7. Statistical charts of key geometric parameters for the lava tube cross-sections.

trial habitats. A comprehensive assessment framework must integrate multiple aspects, including cave identification, structural stability, and geological context (Titus et al., 2021). Quantitative geometric analysis provides an indispensable data foundation and preliminary screening criteria for such integrated evaluations. Recent studies have strongly demonstrated that cross-sectional geometric features, particularly the aspect ratio, are decisive factors in determining structural stability (Wang et al., 2025). Moreover, finite element limit analysis has confirmed that geometric data based on real cross-sectional profiles are essential for analyzing stress distribution and predicting local collapse risks (Chwała et al., 2024). Therefore, the high-precision morphological parameters obtained in this study establish a solid geometric foundation for subsequent numerical simulations and safety assessments of structural stability.

4.3. Resampling and surface reconstruction

4.3.1. Resampling methods comparison and reconstruction impact

The quality of surface reconstruction largely depends on resampling accuracy, as it directly affects the smoothness and local geometric fidelity of the input point cloud. Based on the OSR model, this study optimizes normal vector computation by introducing an adaptive normal mechanism. This mechanism enhances the adaptability of normal estimation across regions with different curvatures while

retaining the original resampling framework. To verify the effectiveness of the proposed method, this paper designed comparative experiments to analyze the quality of different resampling methods and their relationship with the final surface reconstruction accuracy. The experiment selected ten representative feature regions from lava tube point clouds, covering various geometric forms such as flat surfaces, curvature transition areas, and edge features. The initial point clouds of each region were resampled using the ASR method proposed in this paper, the OSR model resampling method, and the MLS resampling method, and the geometric and curvature similarity metrics were computed. The results are shown in Fig. 8 and Fig. 9, respectively.

Research on the OSR model shows that when the radius is 50 mm and the normal deviation threshold is 5°, the resampling accuracy is highest (Yao et al., 2025); therefore, this comparative experiment uses this set of parameters as the benchmark. The radius parameter for MLS is set to 50 mm. The initial neighborhood radius for ASR is set to 50 mm. Resampling results indicate that the ASR outperforms OSR and MLS in both geometric similarity and curvature similarity: its geometric similarity is more than 20% higher than OSR and more than 40% higher than MLS; its curvature similarity is more than 30% higher than OSR and more than 33% higher than MLS. In addition, the ASR demonstrates more stable preservation of geometric and curvature features across different regions, reflecting stronger adaptability and robustness.

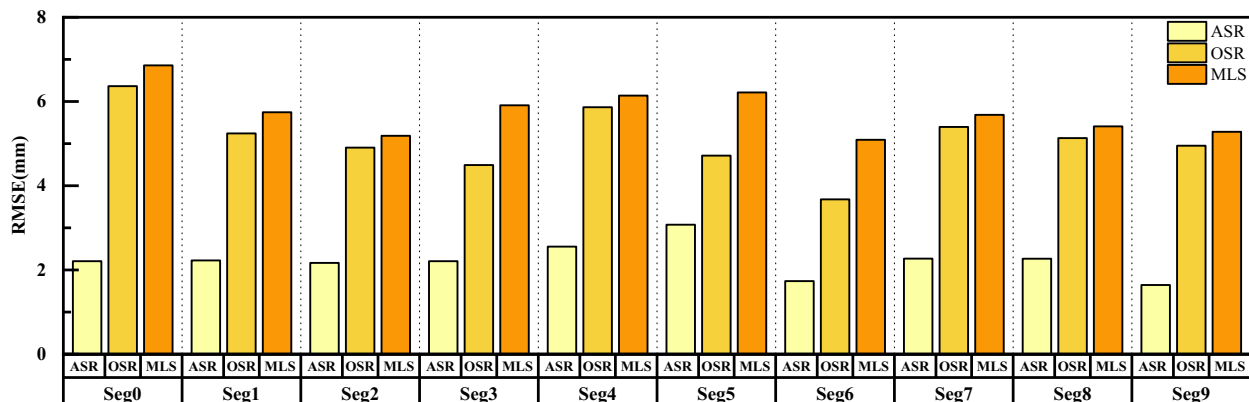


Fig. 8. Resampling methods geometric similarity comparison.

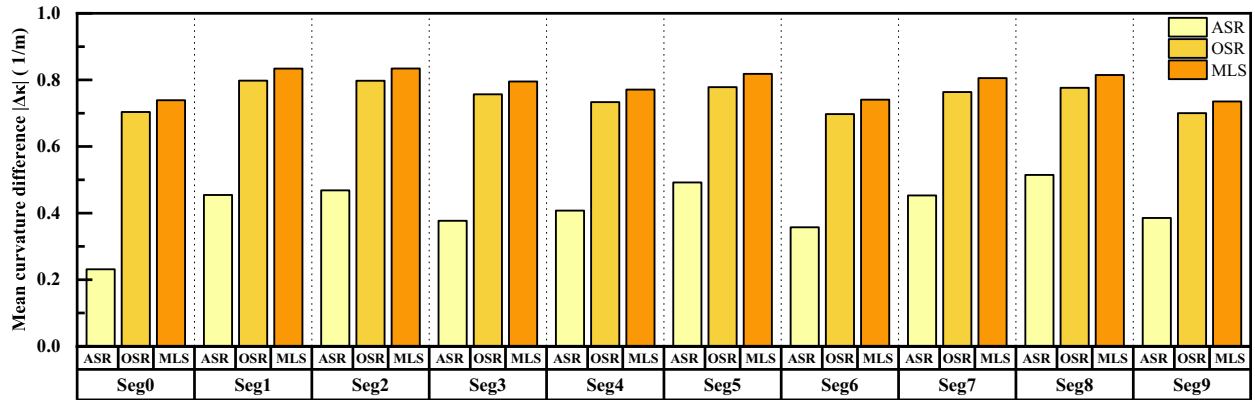
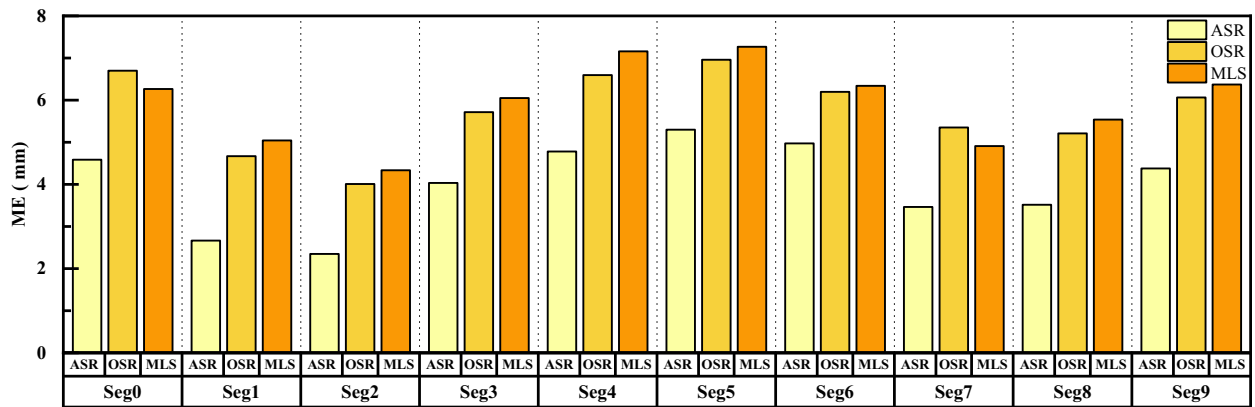
Fig. 9. Mean curvature difference $|\Delta\kappa|$ of resampling methods.

Fig. 10. ME of surfaces reconstructed from point clouds resampled using ASR, OSR, and MLS methods.

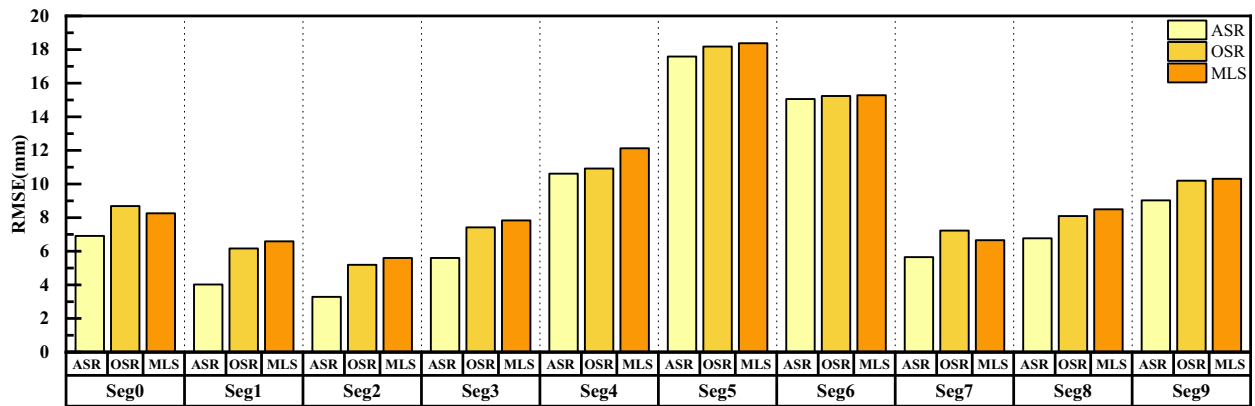


Fig. 11. RMSE of surfaces reconstructed from point clouds resampled using ASR, OSR, and MLS.

On this basis, as described in Section 2.2.4, the implicit Poisson surface reconstruction is performed on the point clouds resampled by the three methods, and the average error (ME) and root mean square error (RMSE) of the reconstructed surfaces are calculated to evaluate their accuracy. The ME and RMSE results are shown in Fig. 10 and Fig. 11. Experimental results indicate that the ASR method achieves the lowest reconstruction errors, with a minimum ME as low as 2 mm. These findings demonstrate that

improving the accuracy of the resampling stage can effectively enhance the quality of the final reconstructed surface.

4.3.2. Parameter analysis of ASR

This section focuses on the impact of the initial neighborhood radius in ASR models on resampling and surface reconstruction. The initial neighborhood radius is a key parameter, as it determines the scale of local points considered in normal vector estimation, directly affecting estima-

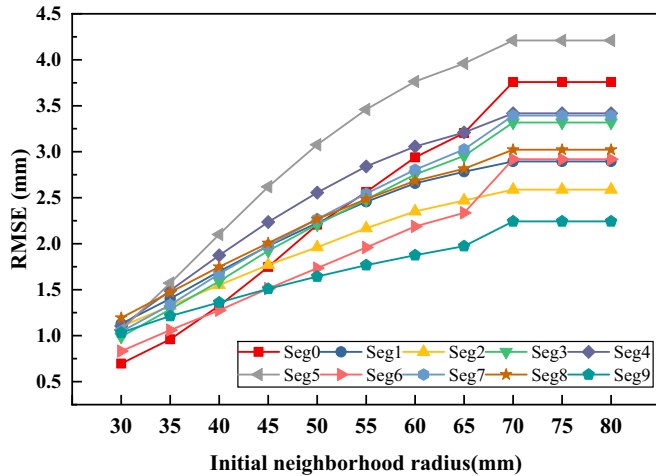


Fig. 12. Geometric similarity of ASR resampling in ten feature regions (Seg0-Seg9) under different initial radius.

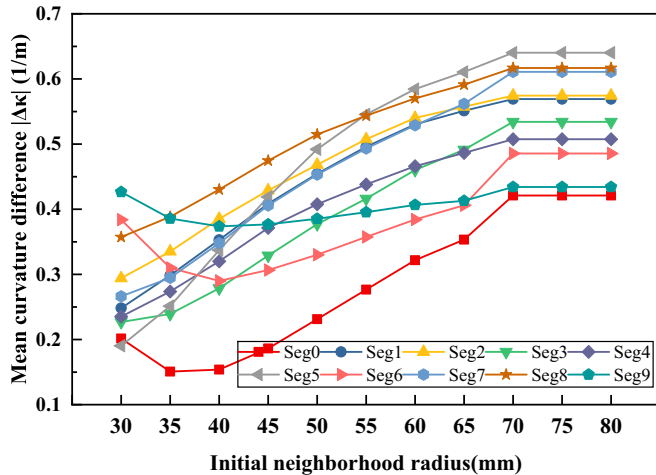


Fig. 13. Mean curvature difference of ASR resampling in ten feature regions (Seg0-Seg9) under different initial radius.

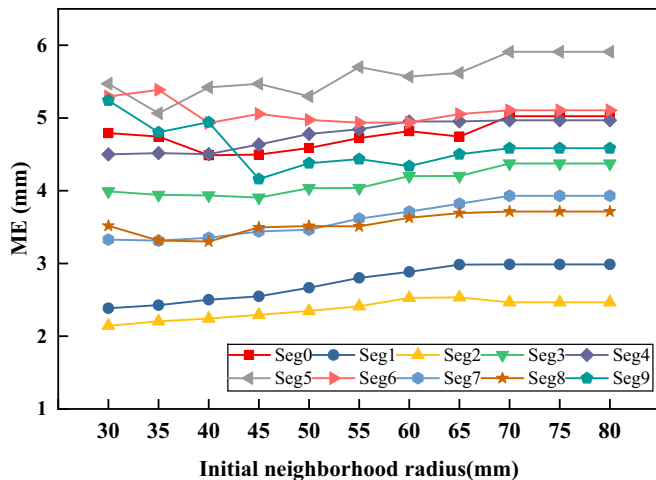


Fig. 14. Mean Error reconstructed surface in ten feature regions (Seg0-Seg9) under different initial radius.

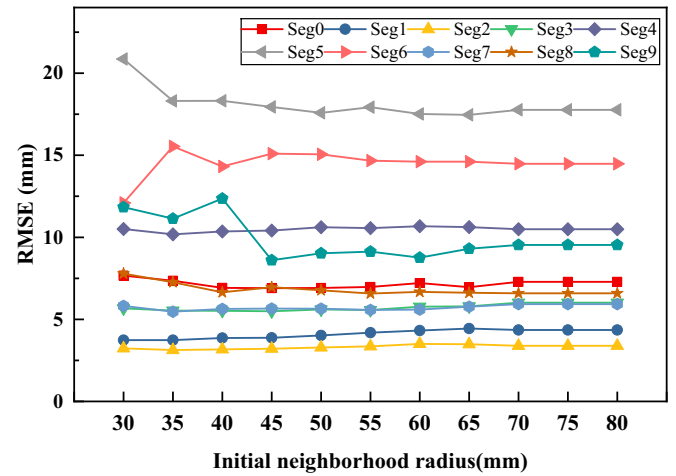


Fig. 15. RMSE reconstructed surface in ten feature regions (Seg0-Seg9) under different initial radius.

tion accuracy and stability. To systematically analyze its influence on resampling quality, this study designed a comparative experiment: the initial neighborhood radius was set to 11 values ranging from 30 mm to 80 mm (30, 35, 40, 45, 50, 55, 60, 65, 70, 75, and 80 mm), and resampling was performed on 10 feature regions, from seg0 to seg9. The geometric similarity and curvature similarity of the point clouds for the feature regions under different initial neighborhood radii are shown in Fig. 12 and Fig. 13, respectively.

Experimental results show that as the initial neighborhood radius increases, the geometric similarity and curvature similarity of ASR resampling decrease. When the initial neighborhood radius exceeds 70 mm, the changes in geometric similarity and curvature similarity tend to stabilize, and the influence of additional points on local features reaches saturation. The trends of geometric similarity and curvature similarity across different feature regions are highly consistent, with no significant region-specific deviations, indicating that the impact of the initial neighborhood radius on resampling quality is robust across various feature regions.

Surface reconstruction was performed on the resampled point clouds, and the MS and RMSE of the reconstructed surfaces were calculated to evaluate reconstruction accuracy. The results, shown in Fig. 14 and Fig. 15, indicate that the reconstruction error tends to stabilize when the initial neighborhood radius exceeds 45 mm.

5. Conclusion

Lava tubes possess significant scientific research value (Blamont, 2014). To evaluate their feasibility as potential extraterrestrial underground bases, it is crucial to investigate their morphological characteristics. This paper proposes a morphological analysis framework that extracts the central axis of lava tubes using a MS algorithm and

computes geometric parameters for 27 representative cross-sections. To meet the high-precision surface reconstruction requirements of morphological studies, a high-precision surface reconstruction method based on curvature optimization is further proposed and validated using the Indian Lava Tube point cloud dataset. This method integrates PCA, MLS and the Poisson implicit surface reconstruction algorithm to perform point cloud resampling and generate high-quality surfaces. In the experiments, the initial neighborhood radius was selected in the range of 30 mm to 80 mm. Ten characteristic regions of the lava tube were resampled, and surface reconstruction was performed. The results show that ASR can achieve surface reconstruction with a minimum RMSE of 3 mm. The method proposed in this study can significantly improve reconstruction accuracy while preserving local structural details. Although validated on terrestrial lava tube data, their geometric complexity effectively simulates extraterrestrial structures. In extraterrestrial missions, limitations of payload, power, and mobility will yield much lower LiDAR point densities than terrestrial scans; however, the ASR method performs adaptive reconstruction based on local geometry and remains robust to density variations. Thus, ASR shows strong potential for high-precision reconstruction under low-density, high-noise conditions, providing theoretical and methodological support for extraterrestrial lava tube characterization and base site selection.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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